

Problem

The purpose of this project was to **create a video game**, specifically in the **turn-based RPG** genre that provides the player with a satisfying experience of creating a character and completing a quest as a first time game dev. Since I had never developed a game, I aimed to learn about the process through doing this.

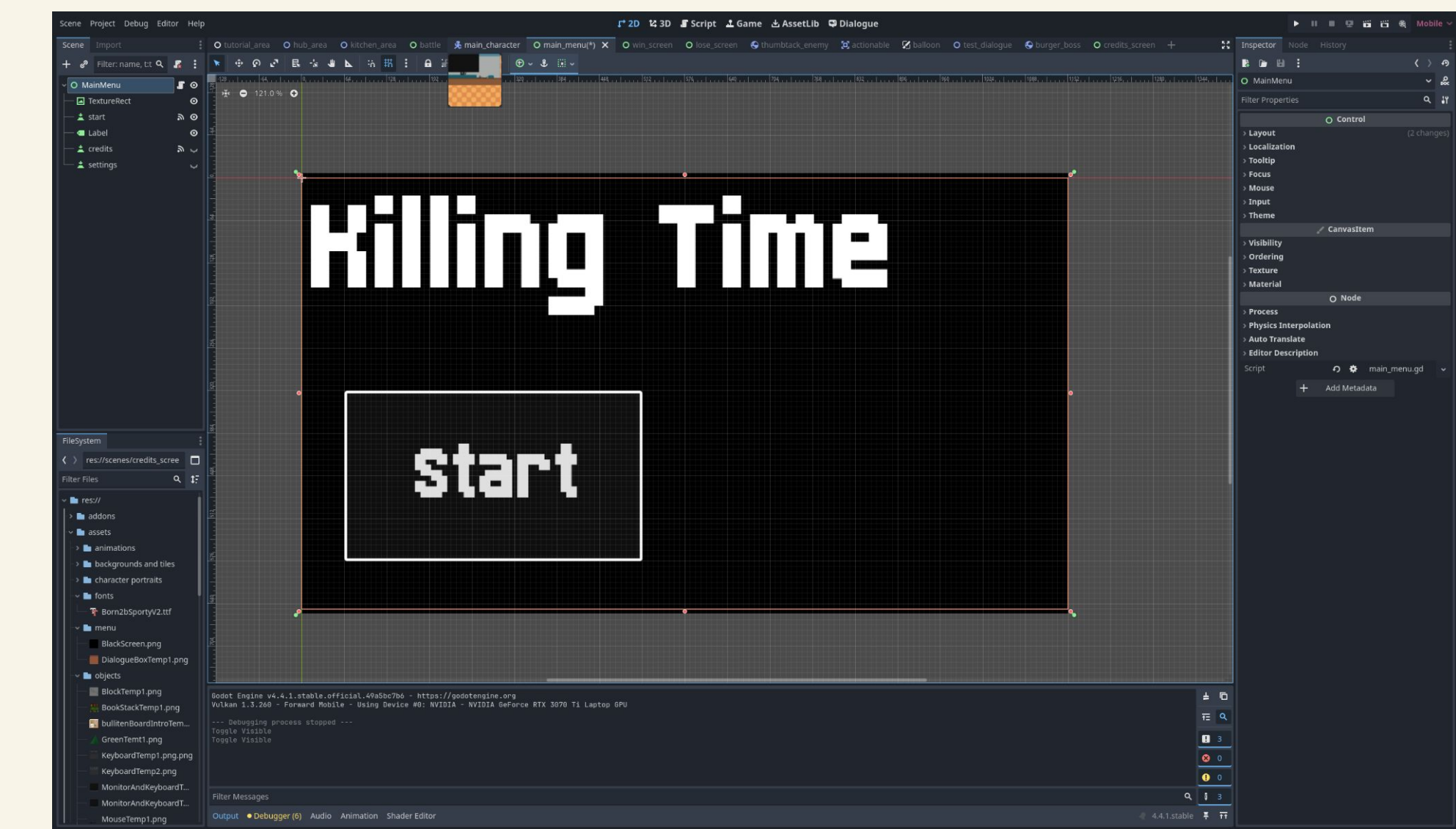
Significance

I created this game because I had **no experience in game development** and **wished to learn** more about the process and gain experience in the field. I hoped to learn everything that goes into developing a game, even if it was a very basic understanding. By dipping my toe into each element of game development, I gave myself the opportunity to see what parts I really enjoyed and what parts I would like to avoid.

Background

A turn-based RPG is a form of **video game** where the player typically makes their own character and **makes decisions** based on who they believe their character is. Alongside this, when the player enters combat, they and the enemy will take turns performing attacks and activating abilities. It's **important for the game to react** to the player's actions in an appropriate way. This is in the sense that if the player performs an attack, the enemy should take damage and react.

Game engines are an important part of making a game. The game engine is the software that a game is built in. **Godot** is an engine that specializes in **2-D games** and has many online resources with tutorials, perfect for a first time game dev making a 2-D game.

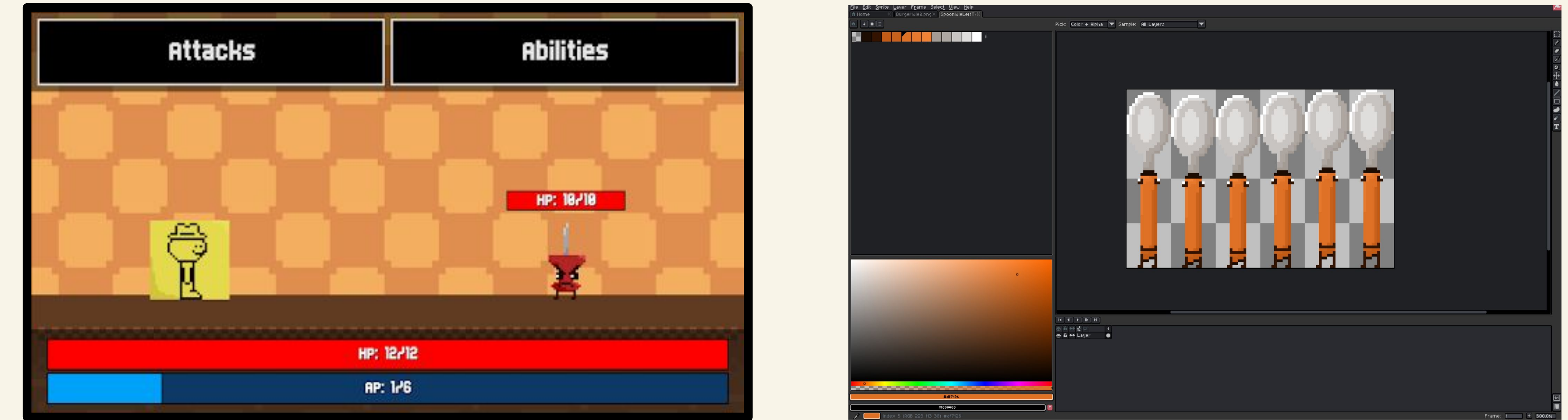


Killing Time: Creation of a Turn-Based 2-D RPG Video Game Within the Godot Engine

Method

The game was built in **sections**. The first section consisted of the first stage that had ground to walk on, a background, and a character to converse with and battle. Following that, additional stages were constructed using the previously existing components alongside a title screen and ending screens. Creating new characters for the levels consisted of creating new assets and plugging in new numbers for health and damage output. This style of design was chosen because it allowed for the player to have a cohesive experience despite the length of the game being short. Throughout the entire process, **art assets would be created alongside the areas and enemies** that were being added.

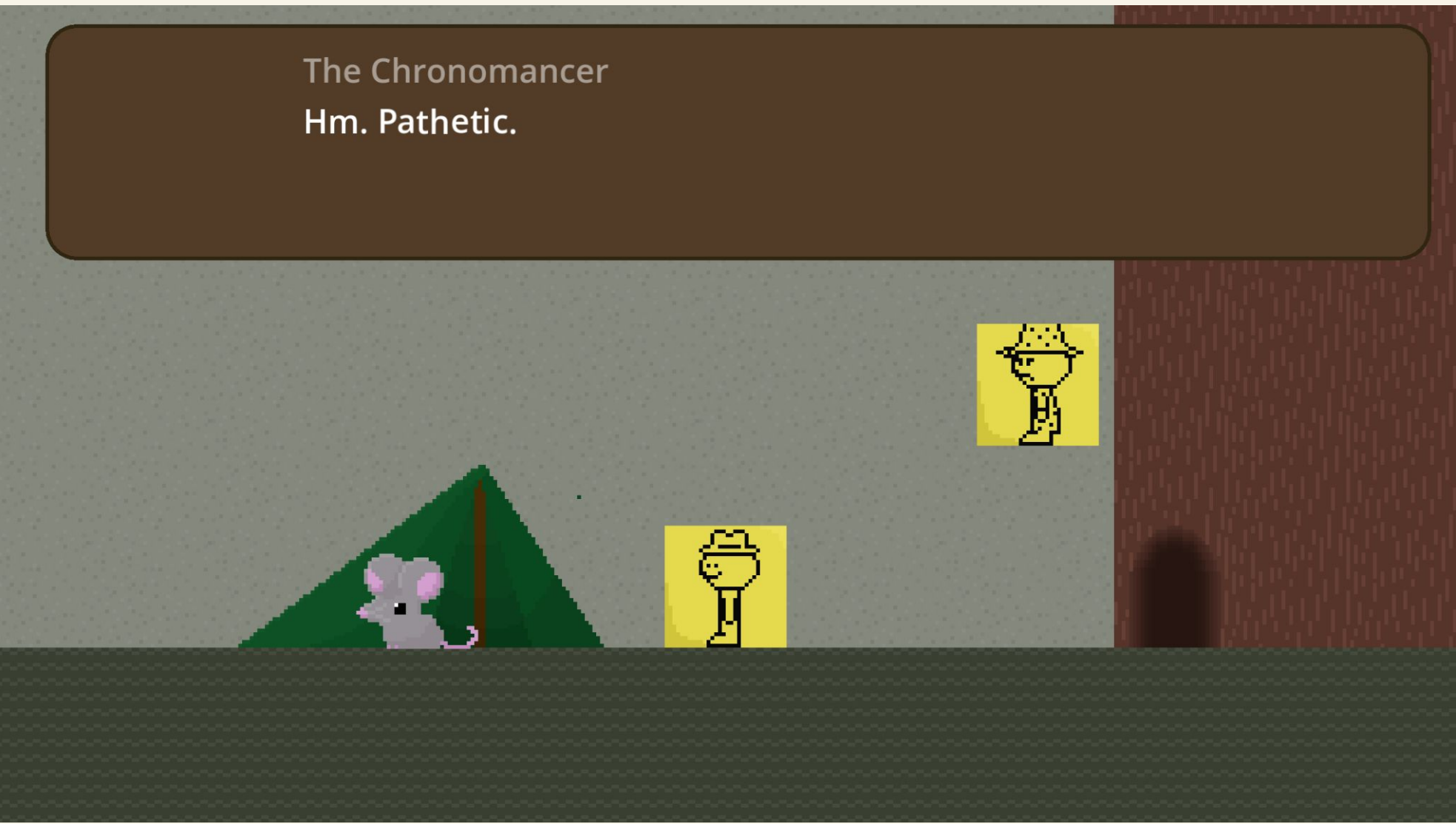
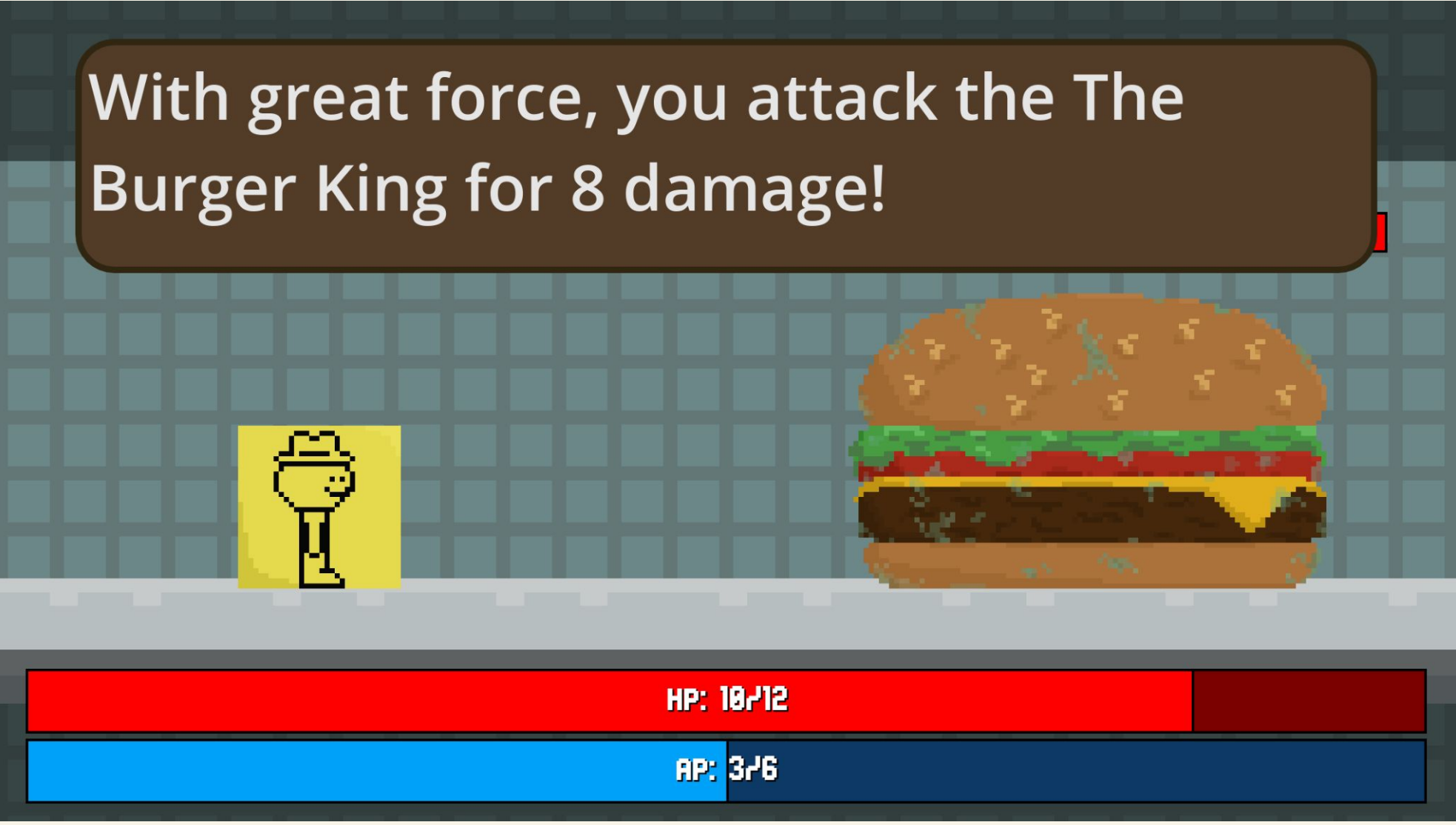
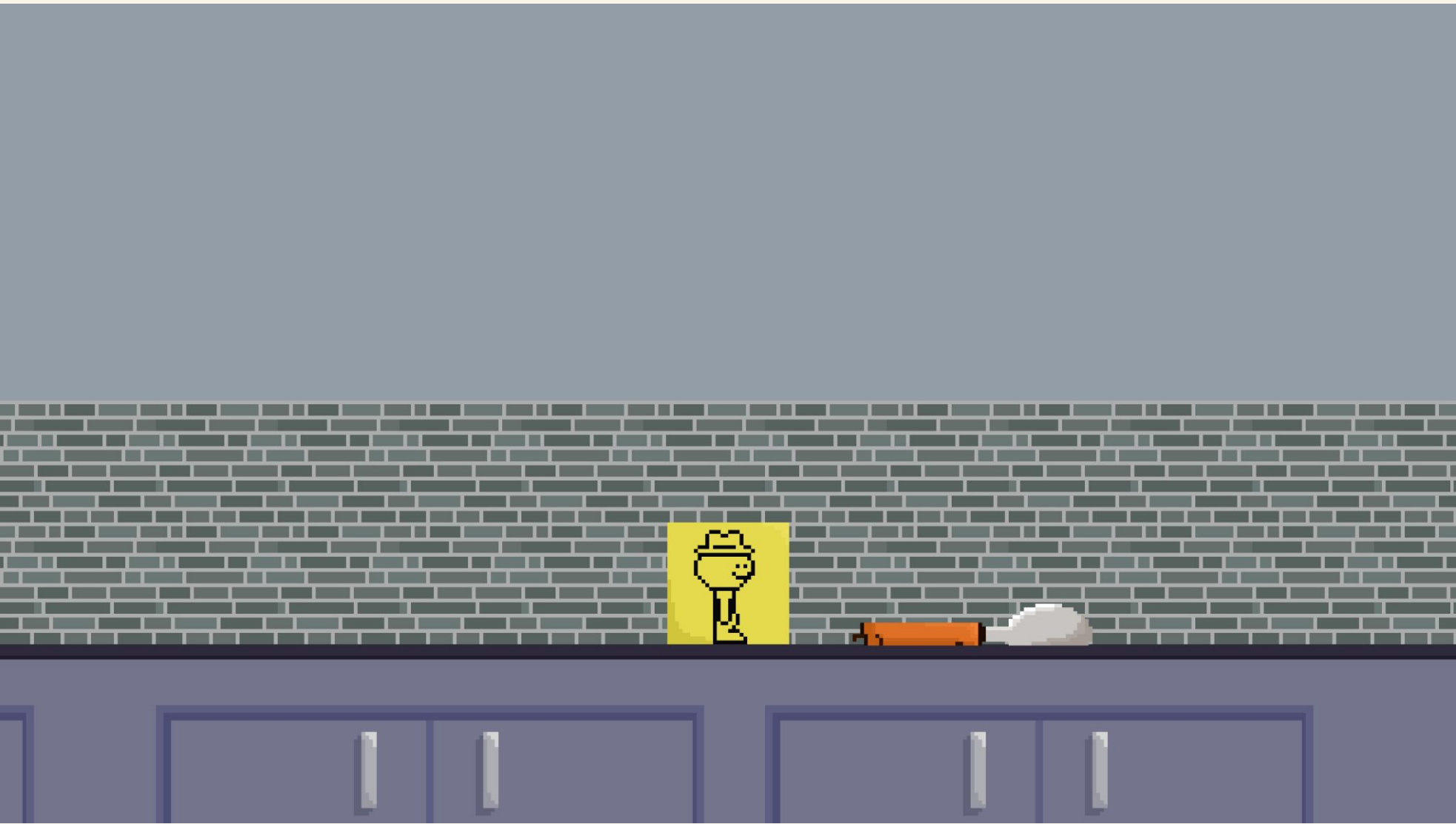
The game **did not have formal testing** because there wasn't any specific data that was looked for. The **game was playtested** for functionality and playability. This occurred throughout the development process and feedback was taken constantly.



Results

The project is in a **presentable state**. The program opens to a title screen that prompts the player to start. Upon starting, the first area that consists of the desk and a portion of the floor beside the desk is finished. The desk screen also has a thumbtack enemy that initiates dialogue and brings the player to a combat screen. **Combat is functional and fairly fun.** The player is provided with **multiple options for actions** to take that vary from dealing damage to preparing defenses to storing action points for upcoming turns. Upon completion of combat, the player can traverse to the next screen where they **acquire a quest** to defeat The Burger King, who resides in the fridge in the breakroom, from a mouse in the wall. The next screen is the breakroom where the player battles one additional enemy before encountering The Burger King. A more difficult fight ensues, and upon completion, the player returns to the quest-giving mouse and an **ending sequence where the chronomancer shows up**. Throughout the entire game, the player is looking at **assets that were completely original creations**.

Because the game is a **cohesive experience** with a beginning, middle, and end, **additional work on the project is not required**; however, it would be improved with additional work put into expanding the scope.



Conclusion

Overall, a **majority of the design goals were met**. The game is a functional 2-D turn-based RPG. There were **some elements** of the game that were originally planned to have been added, but they were **dropped** throughout the process to **prioritize different parts**. For instance, the game was originally planned to have an experience system that would allow the player to level up.

If given more time to continue creating the game, the next step would be to **expand upon the length** of the game. The components are modular enough where adding **new areas** is mostly a matter of creating assets and writing the story. For anyone attempting to recreate a project similar to this one, ensure that enough **time is put towards creating art assets** because they were more time consuming than expected.

Overall, the project was a good experience in learning how to develop a video game solo and a similar experience is a good idea for any up and coming game developers to understand everything that goes into creating a game.